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Andaliman development as a regional superior product in Toba district North Sumatra Province based on regional innovation system

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Abstract. Andaliman as a spice that is closely related to people's lives in Toba Regency can provide added value through a regional innovation system (RIS). This study aims to analyze the development of andaliman as a superior product in the Lake Toba area based on RIS. The researchwas conducted in October 2021, located in Toba Regency, North Sumatra Province. Key informants in the research are: Regional Development Planning Agency; Department of Agriculture and Fisheries; Department of Trade, Industry and Small and Medium Enterprise Cooperatives; andaliman farmers; as well as, entrepreneurs of andaliman processed products. This study uses a qualitative method with an observation approach. Data were collected through: documentation studies, surveys, interviews and Focus Group Discussion. The development of andaliman as a regional superior product based on the regional innovation system in Toba Regency has the potential to be realized with the support of innovation actors consisting of academic, business, community, and government components.

1. Introduction

Andaliman is a typical plant of North Sumatra and is closely related to the life of the Batak people wholive around the Lake Toba area. Andaliman is included in the type of spice with the Latin name *Zanthoxylum acanthopodium* DC and is a superior commodity in the Lake Toba area [1][2]. Andaliman seeds are commonly used in various traditional Batak dishes such as arsik, natinombur, naniura and sambal Tuktuk [3]. While the leaves are commonly called the *sinyar-sinyar* leaves by the local community and are useful for treating stomach aches [4]. Currently andaliman is not only used as a spice in traditional cuisine, various modern confectionery has used andaliman as one of its compositions such as pizza, chips, chili sauce, coffee, bandrek and sweets. Andaliman also has the potential as a producer of essential oils, and can contribute to the world of health because it produces compounds that can function as anti-bacterial, anti-oxidant, anti-diabetic, anti-cancer, anti-inflammatory, anti-aging, anti-acne, anti- fertility, and anti-osteoporosis [5].

Andaliman belongs to the Rutaceae family and grows in other countries such as India, China and Tibet. The shape is like pepper, small round, green in color, and will blacken when it dries [6]. Andaliman is a shrub, erect, with a height of up to 5 meters. Andaliman seed germination occurred above ground with low germination rate of 41%. So that the age of andaliman germination is relatively long and varies, ranging from 24 to 100 days after sowing. This condition is due to the

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tough structure of the seed coat which prevents the imbibition of water and gas exchange in the germination process. The majority of Andaliman grown in North Sumatra grows naturally from seeds dispersed by birds, as well as seeds obtained by farmers accidentally from burning weeds in areas of mature plants [2].

The various benefits of andaliman make this product have the potential to be widely developed. In addition to increasing production in the form of raw materials, it is necessary to develop a system that has an impact on the development of Andaliman into products that have added value. Abundant naturalresources will have a positive impact on the regional economy if they are managed and provide added value. As has been done by Thailand as the world's largest rice exporter, which seeks to increase the added value of rice through the development of downstream product innovations from rice [7]. The role of innovation in increasing the added value of products and regional competitiveness must be supported by actors who play a role in carrying out innovation. As the results of research in the outskirts of Canada which concluded that the potential for innovation can grow rapidly if there is government support for the role of public organizations as drivers of the innovation process and economic change [8].

The concept of innovation in general aims to understand how a set of institutions, organizations, networks, and actors can interact to encourage innovation on a national, regional, sectoral, or laboratoryscale built by technology development companies [9]. The Indonesian government has addressed the importance of innovation by issuing several policies, namely: Joint Regulation of the Minister of Research and Technology and the Minister of Home Affairs Number 03 of 2021 and number 36 of 2012concerning strengthening regional innovation systems; Government Regulation Number 38 of 2017 concerning regional innovation.

Refer to the Joint Regulation of the Minister of Research and Technology and the Minister of HomeAffairs Number 03 of 2021 and number 36 of 2012, regional innovation system institutions consist of: government institutions, regional governments, research and development institutions, educational institutions, innovation support institutions, the business world, and community organizations [10]. Furthermore, these institutions are grouped into 4 actors, namely: Academics (A), Business (B), Community (C), and Government (G).

Based on this background, this study aims to analyze the development of andaliman as a superior product in the Lake Toba area based on RIS. This research will provide an overview of the RIS institutions, especially for andaliman development so that they have added value. So that it has an impact on increasing regional competitiveness and people's income.

2. Materials and methods

2.1. Location and samples

The research was conducted in October 2021, located in Toba Regency, North Sumatra Province. Key informants in the research are: Regional Development Planning Agency; Department of Agriculture and Fisheries; Department of Trade, Industry and Small and Medium Enterprise Cooperatives; andaliman farmers; as well as, entrepreneurs of andaliman processed products.

2.2. Analysis method

This study uses a qualitative method with an observation approach. Data were collected through: documentation studies, surveys, and interviews [11]. Furthermore, a Focus Group Discussion was conducted with all research informants to formulate the roles and responsibilities of each actor in the development of andaliman as a superior product. Primary data consists of: andaliman planting area, business units and types of andaliman processed products, number of entrepreneurs. Analysis of andaliman development as a regional superior product is carried out by mapping the role of each actor according to the concept of the regional innovation system.

3. Results and discussion

3.1. Land area and production

The distribution of Andaliman agricultural land in Toba Regency is in 4 sub-districts, namely: Borbor, Habinsaran, Lumban Julu, and Silaen. The quantity of andaliman production in each sub-district is presented in Table 1. Since 2018, land area and quantity of andaliman production have continued to experience as presented in Table 2. In total, andaliman production in Toba Regency has reached 300 tons per year.

Table 1. Andaliman planting area.

No.	Sub-district	Area (ha)
1.	Borbor	103.28
2.	Habinsaran	141.18
3.	Lumban Julu	No data
4.	Silaen	No data

Source: Research data (2021)

Table 2. Andaliman production.

Year	Production (Ton)	Area (ha)
2017	No data	60.00
2018	14.00	60.00
2019	12.90	66.02
2020	45.00	133.00

Source: Research data (2021)

Based on the information from Department of Agriculture and Fisheries of Toba Regency, data of the amount of production and land area are not always available because andaliman cultivation has not been actively carried out by the community, because andaliman seed certification is on processing stage at the Ministry of Agriculture. The majority of andaliman harvested by farmers are wild plants that grow in fields or new openings in the forest, so their distribution is uneven.

However, a practitioner has cultivated andaliman in his working area called Taman Eden 100. Cultivation is done through seeds and shoot cuttings. Another problem is the fluctuating price of fresh andaliman among farmers. Occasionally farmers experience losses while andaliman is abundant in the market. The selling price is very cheap with a price range between idr 40,000 to idr 50,000 per kg. Given that fresh andaliman cannot last long, farmers must sell it at a low price or it will rot.

3.2. Andaliman processed products

MSME entrepreneurs in Toba Regency have used andaliman as a raw material or mixture for various products, such as: dried andaliman seeds, andaliman powder, andaliman chili sauce, andaliman bandrek,andaliman chips, andaliman drinks, andaliman sasagon, and andaliman pizza. Some products already have distribution permits, while the marketing is still local. Figure 1 presents andaliman processed products that have been produced by the community

Figure 1. Andaliman processed products.

3.3. Regional innovation system actors and their roles

The application of the regional innovation system for the development of andaliman can be done by identifying the actors who play a role. Furthermore, describing the problem, as described on the previous two sub-chapters. Based on the results of research analysis, the actors who play a role in the development of andaliman based on regional innovation systems are presented in Table 3. The role of each actor in general is to support the development of andaliman as a regional superior product in Toba Regency. Actors in the regional innovation system are then gathered in the coordination team.

Table 3. Regional innovation system (RIS) actor.

No.	RIS Actor	Person In Charge
1.	Academics	Institut Teknologi Del, Simalungun University
2.	Business	Taman Eden 100; Toba Caldera MSME Association
3.	Community	Farmer; MSME entrepreneurs;
4.	Government	Regional Planning and Development Agency; Department of Agriculture and Fisheries; Department of Trade, Industry and Small and Medium Enterprise Cooperatives.

The Coordination Team is a forum for building commitment among RIS actors. The regional head acts as a guide in the Coordination Team and ensures policies for strengthening RIS in the Regency, while the Regional Secretary acts as the Chair for strengthening RIS. The RIS policy is prepared by theCoordination Team and produces a roadmap document for strengthening RIS, which is then integrated into the District Medium Term Development Plan (RPJMD). Relevant agencies that are included in theRIS actors are responsible for ensuring that every activity in the RIS roadmap is listed in the RegionalApparatus Activity Plan (RKPD).

3.4. Andaliman development activity plan based on RIS

The plan for the development of andaliman as a regional superior product is in accordance with the Toba Regency's 4th mission, namely to spur quality and competitive economic growth, and the 5th mission, namely to improve natural resource management and agriculture that is oriented towards environmental sustainability and community welfare. The activity plan must also address problems in the development of andaliman as a regional superior product.

Table 4. Objective and activity plan of RIS.

Objective	Activity Plan	Person In Charge
Improve the management of regional superior products	Andaliman-based entrepreneurship training	Institut Teknologi Del, Simalungun University; Department of Trade, Industry and Small and Medium Enterprise Cooperatives; MSME entrepreneurs; Toba Caldera MSME Association
	Andaliman processed product processing training	MSME entrepreneurs; Toba Caldera MSME Association; Department of Trade, Industry and Small and Medium Enterprise Cooperatives
	Digital marketing training	Institut Teknologi Del; MSME entrepreneurs; Toba Caldera MSME Association; Department of Trade, Industry and Small and Medium Enterprise Cooperatives
	Training on product packaging	MSME entrepreneurs; Toba Caldera MSME Association; Department of Trade, Industry and Small and Medium Enterprise Cooperatives
	Technical Guidance for business licensing, such as: PIRT, Halal, BPOM, HACCP	Department of Trade, Industry and Small and Medium Enterprise Cooperatives
Increase agricultural production and productivity	Andaliman cultivation training	Simalungun University; Taman Eden 100; Farmer; Department of Agriculture and Fisheries
	Pest prevention training on andaliman plants	Simalungun University; Taman Eden 100; Farmer; Department of Agriculture and Fisheries

Table 4 describes the planned activities that can be carried out in order to develop andaliman as a regional superior product. The activity plan suggested in Table 4 focuses more on individual development as an actor of innovation in the development of andaliman as a superior product. This is inline with research [8] that increasing individual competence will support the organization and the innovation system that has been built. Collaboration between actors in the implementation of activities is the key to the success and sustainability of regional innovation systems, such as the results of research [7] which concludes that collaboration between institutions will have an impact on more formal relationships, connectedness of activities and recognition of products produced by regional innovation systems.

4. Conclusions

Based on the research results, the Andaliman commodity in Toba Regency has the potential to be developed based on RIS. In order to encourage the development of Andaliman products, the action plansthat must be carried out are: Andaliman-based entrepreneurship training; Andaliman processed product processing training; Digital marketing training; Training on product packaging; Technical Guidance forbusiness licensing, such as: PIRT, Halal, BPOM, HACCP; Andaliman cultivation training; and, Pest prevention training onandaliman plants.

Recommendations for the development of andaliman based on the RIS are to the regional development planning agency to strengthen innovation actors (academic, business, community, government) and create a road map for the development of andaliman as a regional superior product based on the RIS together with the RIS coordination team.

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